Andrea Celli

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Employment

Assistant Professor — **Milan**
Bocconi University, Computing Sciences Department — *September 2021 – Present*

Postdoctoral researcher — **London**
Facebook Core Data Science — *June 2020 – June 2021*

Postdoctoral researcher — **Milan**
Politecnico di Milano — *November 2019 – May 2020*

Other Affiliations

Research Affiliate
Bocconi Institute for Data Science and Analytics — *September 2021 – Present*

Member
European Laboratory for Learning and Intelligent Systems (ELLIS) — *October 2021 – Present*

Education

Politecnico di Milano — **Milan**
PhD in Computer Science — *2016 – February 2020*

Carnegie Mellon University — **Pittsburgh**
Visiting Scholar — *Jan 2018 – May 2018*

Politecnico di Milano — **Milan**
MSc in Computer Science and Engineering, Grade: 110 cum laude/110 — *2014 – 2016*

Politecnico di Milano — **Milan**
BSc in Computer Science and Engineering, Grade: 110 cum laude/110 — *2011 – 2014*

Publications

Conference Papers

[C1] *The Complexity of Min-Max Optimization for Quadratic Polynomials*
M. Bernasconi, M. Castiglioni, A. Celli, and A. Hollender, *SODA*, 2027.

[C2] *Min-Max Optimization Requires Exponentially Many Queries*
M. Bernasconi, M. Castiglioni, A. Celli, and A. Hollender, *NeurIPS*, 2026.

[C3] *Regret Minimization in Single-Dimensional Contract-Design with Binary Actions*
R. Poiani, M. Bernasconi, and A. Celli, *NeurIPS*, 2026.

[C4] *A Unifying Framework for Quasi-Polynomial Optimization of Fixed-degree Polynomials*
M. Bernasconi, M. Castiglioni, A. Celli, and G. Farina, *FOCS*, 2026.

[C5] *Improved Hardness Results for Min-Max Optimization with Coupled Constraints*
M. Bernasconi, M. Castiglioni, A. Celli, and G. Farina, *EC*, 2026.

[C6] *Optimal Rates for Feasible Payoff Set Estimation in Games*
A. Barbara, R. Poiani, M. Bernasconi, and A. Celli, *ICML (Spotlight)*, 2026.

[C7] *Pure Exploration with Infinite Answers*
R. Poiani, M. Bernasconi, and A. Celli, *AISTATS (Spotlight)*, 2026.

[C8] *Nonparametric Contextual Online Bilateral Trade*
E. Coccia, M. Bernasconi, and A. Celli, *ICLR*, 2026.

[C9] *Non-Asymptotic Analysis of (Sticky) Track-and-Stop*
R. Poiani, M. Bernasconi, and A. Celli, *ICLR (Oral)*, 2026.

[C10] *Nearly-optimal bandit learning in stackelberg games with side information*
M.-F. Balcan, M. Bernasconi, M. Castiglioni, A. Celli, K. Harris, and Z. S. Wu, *ICLR*, 2026.

[C11] *The Complexity of Correlated Equilibria in Generalized Games*
M. Bernasconi, M. Castiglioni, A. Celli, and G. Farina, *NeurIPS*, 2025.

[C12] *No-regret is not enough! Bandits with general constraints through adaptive regret minimization*
M. Bernasconi, M. Castiglioni, and A. Celli, *ICML*, 2025.

[C13] *Online Learning in the Random-Order Model*
M. Bernasconi, A. Celli, R. Colini-Baldeschi, F. Fusco, S. Leonardi, and M. Russo, *ICML*, 2025.

[C14] *Feature-based online bilateral trade*
S. Gaucher, M. Bernasconi, M. Castiglioni, A. Celli, and V. Perchet, *ICLR*, 2025.

[C15] *Beyond primal-dual methods in bandits with stochastic and adversarial constraints*
M. Bernasconi, M. Castiglioni, A. Celli, and F. Fusco, *NeurIPS*, 2024.

[C16] *Online learning with sublinear best-action queries*
M. Russo, A. Celli, R. Colini Baldeschi, F. Fusco, D. Haimovich, D. Karamshuk, S. Leonardi, and N. Tax, *NeurIPS*, 2024.

[C17] *Bandits with Knapsacks and Predictions*
D. Drago, A. Celli, and M. Elias, *UAI*, 2024.

[C18] *Online Learning Under Budget and ROI Constraints via Weak Adaptivity*
M. Castiglioni, A. Celli, and C. Kroer, *ICML*, 2024.

[C19] *No-Regret Learning in Bilateral Trade via Global Budget Balance*
M. Bernasconi, M. Castiglioni, A. Celli, and F. Fusco, *STOC*, 2024.

[C20] *Agent-Designed Contracts: How to Sell Hidden Actions*
M. Bernasconi, M. Castiglioni, and A. Celli, *EC*, 2024.

[C21] *Steering No-Regret Learners to a Desired Equilibrium*
B. Zhang, G. Farina, I. Anagnostides, F. Cacciamani, S. McAleer, A. Haupt, A. Celli, N. Gatti, V. Conitzer, and T. Sandholm, *EC*, 2024.

[C22] *Bandits with Replenishable Knapsacks: the Best of both Worlds*
M. Bernasconi, M. Castiglioni, A. Celli, and F. Fusco, *ICLR*, 2024.

[C23] *Computing optimal equilibria and mechanisms via learning in zero-sum extensive-form games*
B. Zhang, G. Farina, I. Anagnostides, F. Cacciamani, S. McAleer, A. Haupt, A. Celli, N. Gatti, V. Conitzer, and T. Sandholm, *NeurIPS*, 2023.

[C24] *Optimal Rates and Efficient Algorithms for Online Bayesian Persuasion*
M. Bernasconi, M. Castiglioni, A. Celli, A. Marchesi, F. Trovò, and N. Gatti, *ICML*, 2023.

[C25] *Fully Dynamic Online Selection through Online Contention Resolution Schemes*
V. Avadhanula, A. Celli, R. Colini-Baldeschi, S. Leonardi, and M. Russo, *AAAI*, 2023.

[C26] *A Unifying Framework for Online Optimization with Long-Term Constraints*
M. Castiglioni, A. Celli, A. Marchesi, G. Romano, and N. Gatti, *NeurIPS*, 2022.

[C27] *Online Learning with Knapsacks: the Best of Both Worlds*
M. Castiglioni, A. Celli, and C. Kroer, *ICML (Spotlight)*, 2022.

[C28] *Optimal Correlated Equilibria in General-Sum Extensive-Form Games: Fixed-Parameter Algorithms, Hardness, and Two-Sided Column-Generation*
B. Zhang, G. Farina, A. Celli, and T. Sandholm, *EC*, 2022.

[C29] *Faster No-Regret Learning Dynamics for Extensive-Form Correlated and Coarse Correlated Equilibria*
I. Anagnostides, G. Farina, C. Kroer, A. Celli, and T. Sandholm, *EC*, 2022.

[C30] *Fair Equilibria in Sponsored Search Auctions: the Advertisers' Perspective*
G. Birmpas, A. Celli, R. Colini-Baldeschi, and S. Leonardi, *IJCAI*, 2022.

[C31] *The parity ray regularizer for pacing in auction markets*
A. Celli, R. Colini-Baldeschi, C. Kroer, and E. Sodomka, *TheWebConf*, 2022.

[C32] *Connecting Optimal Ex-Ante Collusion in Teams to Extensive-Form Correlation: Faster Algorithms and Positive Complexity Results*
G. Farina, A. Celli, N. Gatti, and T. Sandholm, *ICML (Spotlight)*, 2021.

[C33] *Multi-Receiver Online Bayesian Persuasion*
M. Castiglioni, A. Marchesi, A. Celli, and N. Gatti, *ICML*, 2021.

[C34] *Signaling in Bayesian Network Congestion Games: the Subtle Power of Symmetry*
M. Castiglioni, A. Celli, A. Marchesi, and N. Gatti, *AAAI*, 2021.

[C35] *Multi-Agent Coordination in Adversarial Environments through Signal Mediated Strategies*
F. Cacciamani, A. Celli, M. Ciccone, and N. Gatti, *AAMAS*, 2021.

[C36] *No-Regret Learning Dynamics for Extensive-Form Correlated Equilibrium*
A. Celli, A. Marchesi, G. Farina, and N. Gatti, *NeurIPS (Oral presentation and best paper award), also appeared at Highlights Beyond EC '21 and IJCAI SCBP '21*, 2020.

[C37] *Online Bayesian Persuasion*
M. Castiglioni, A. Celli, A. Marchesi, and N. Gatti, *NeurIPS (Spotlight)*, 2020.

[C38] *Private Bayesian Persuasion with Sequential Games*
A. Celli, S. Coniglio, and N. Gatti, *AAAI*, 2020.

[C39] *Persuading Voters: It's Easy to Whisper, It's Hard to Speak Loud*
M. Castiglioni, A. Celli, and N. Gatti, *AAAI*, 2020.

[C40] *Learning to Correlate in Multi-Player General-Sum Sequential Games*
A. Celli, A. Marchesi, T. Bianchi, and N. Gatti, *NeurIPS*, 2019.

[C41] *Computing Optimal Ex Ante Correlated Equilibria in Two-Player Sequential Games*
A. Celli, S. Coniglio, and N. Gatti, *AAMAS*, 2019.

[C42] *Personality-Based Representations of Imperfect-Recall Games*
A. Celli, G. Romano, and N. Gatti, *AAMAS (Extended Abstract)*, 2019.

[C43] *Ex Ante Coordination and Collusion in Zero-Sum Multi-Player Extensive-Form Games*
G. Farina, A. Celli, N. Gatti, and T. Sandholm, *NeurIPS*, 2018.

[C44] *Computational Results for Extensive-Form Adversarial Team Games*
A. Celli and N. Gatti, *AAAI*, 2018.

[C45] *On the Complexity of Nash Equilibrium Reoptimization*
A. Celli, A. Marchesi, and N. Gatti, *UAI*, 2017.

[C46] *Team-maxmin equilibrium: efficiency bounds and algorithms*
N. Basilico, A. Celli, G. De Nittis, and N. Gatti, *AAAI*, 2017.

[C47] *Coordinating multiple defensive resources in patrolling games with alarm systems*
N. Basilico, A. Celli, G. D. Nittis, and N. Gatti, *AAMAS*, 2017.

Journal Papers

[J1] *No-Regret Learning in Bilateral Trade via Global Budget Balance*
M. Bernasconi, M. Castiglioni, A. Celli, and F. Fusco, *SIAM Journal on Computing (SICOMP)*, (to appear).

[J2] *Optimal Correlated Equilibria in General-Sum Extensive-Form Games: Fixed-Parameter Algorithms, Hardness, and Two-Sided Column-Generation*
B. H. Zhang, G. Farina, A. Celli, and T. Sandholm, *Mathematics of Operations Research*, 2025.

[J3] *Public Bayesian Persuasion: Being Almost Optimal and Almost Persuasive*
M. Castiglioni, A. Celli, and N. Gatti, *Algorithmica*, 2023.

[J4] *Regret Minimization in Online Bayesian Persuasion: Handling Adversarial Receiver's Types under Full and Partial Feedback Models*
M. Castiglioni, A. Celli, A. Marchesi, and N. Gatti, *Artificial Intelligence Journal (AIJ)*, 2023.

[J5] *Simple Uncoupled No-Regret Learning Dynamics for Extensive-Form Correlated Equilibrium*
G. Farina, A. Celli, A. Marchesi, and N. Gatti, *Journal of the ACM*, 2022.

Working Papers

[C1] *A Subsampling Theorem for Constraint Satisfaction Problems with Large Arity*
M. Bernasconi, M. Castiglioni, A. Celli, G. Farina, and G. Malavolta, *arXiv preprint arXiv:2608.00203*, 2026.

[C2] *Theoretical Foundations of max@k Reinforcement Learning*
R. Poiani, M. Bernasconi, and A. Celli, *arXiv preprint arXiv:2607.17823*, 2026.

[C3] *Single-dimensional Contract Design: Efficient Algorithms and Learning*
M. Bernasconi, M. Castiglioni, and A. Celli, *arXiv preprint arXiv:2502.11661*, 2025.

Refereed Workshop Papers

[W1] *No-Regret Learning in Bilateral Trade via Global Budget Balance*
M. Bernasconi, M. Castiglioni, A. Celli, and F. Fusco, *Ninth Marketplace Innovation Workshop (MIW)*, 2024.

[W2] *A Unifying Framework for Online Optimization with Long-Term Constraints*
M. Castiglioni, A. Celli, A. Marchesi, G. Romano, and N. Gatti, *NeurIPS ML Safety workshop*, 2022.

[W3] *Optimal Correlated Equilibria in General-Sum Extensive-Form Games: Fixed-Parameter Algorithms, Hardness, and Two-Sided Column-Generation*
B. Zhang, G. Farina, A. Celli, and T. Sandholm, *Workshop on Gamification and Multiagent Solutions*, 2022.

[W4] *Connecting Optimal Ex-Ante Collusion in Teams to Extensive-Form Correlation: Faster Algorithms and Positive Complexity Results*
G. Farina, A. Celli, N. Gatti, and T. Sandholm, *Workshop on Reinforcement Learning in Games*, 2022.

[W5] *The Parity Ray Regularizer for Pacing in Auction Markets*
A. Celli, R. Colini-Baldeschi, C. Kroer, and E. Sodomka, *Sixth Marketplace Innovation Workshop (MIW)*, 2021.

[W6] *Faster Algorithms for Optimal Ex-Ante Coordinated Collusive Strategies in Extensive-Form Zero-Sum Games*
G. Farina, A. Celli, N. Gatti, and T. Sandholm, *Workshop on Reinforcement Learning in Games*, 2021.

[W7] *Online Bayesian Persuasion*
M. Castiglioni, A. Celli, A. Marchesi, and N. Gatti, *Workshop on Reinforcement Learning in Games*, 2021.

[W8] *No-Regret Learning Dynamics for Extensive-Form Correlated Equilibrium*
A. Celli, A. Marchesi, G. Farina, and N. Gatti, *Workshop on Reinforcement Learning in Games*, 2021.

[W9] *No-Regret Learning Dynamics for Extensive-Form Correlated Equilibrium*
A. Celli, A. Marchesi, G. Farina, and N. Gatti, *Cooperative AI Workshop*, 2020.

[W10] *Multi-Agent Coordination through Signal Mediated Strategies*
F. Cacciamani, A. Celli, M. Ciccone, and N. Gatti, *Cooperative AI Workshop*, 2020.

[W11] *Faster Algorithms for Optimal Ex-Ante Coordinated Collusive Strategies in Extensive-Form Zero-Sum Games*
G. Farina, A. Celli, N. Gatti, and T. Sandholm, *Cooperative AI Workshop*, 2020.

[W12] *Learning to Correlate in Multi-Player General-Sum Sequential Games*
A. Celli, A. Marchesi, T. Bianchi, and N. Gatti, *Smooth Games Optimization and Machine Learning Workshop*, 2019.

[W13] *Ex ante coordination in team games*
G. Farina, A. Celli, N. Gatti, and T. Sandholm, *Workshop on Reinforcement Learning in Games*, 2019.

[W14] *Nash Equilibrium Reoptimization is Hard*
A. Celli and A. Marchesi, *Algorithmic Game Theory Workshop*, 2017.

[W15] *Coordinating multiple defensive resources in patrolling games with alarm systems*
N. Basilico, A. Celli, G. D. Nittis, and N. Gatti, *Workshop on Adversarial Reasoning in Multi-Agent Systems*, 2017.

External Funding

- **2024-2029**: ERC Starting Grant Project "PLA-STEER". Sole PI. €1M+
- **2023-2025**: MUR PRIN Project "Targeted Learning Dynamics: Computing Efficient and Fair Equilibria through No-Regret Algorithms". Lead PI. €195,200 (my share: €97,600)
- **2022-2023**: "Multi-agent learning and equilibrium (EquiLearn)" funded by Civica Research. With Galit Ashkenazi-Golan, Katerina Papadaki, Bernhard von Stengel (PI), and Mark Voorneveld.

Honors and Awards

- Marco Somalvico Award (awarded by Italian Association for Artificial Intelligence AIxIA) 2025
- Best Italian Young Researcher in Theoretical Computer Science (awarded by the Italian Chapter of EATCS) 2025
- NeurIPS 2020 Best Paper Award 2020
- Lesmo Award for the best MSc Thesis in Artificial Intelligence (awarded by the Italian Association for Artificial Intelligence AIxIA) 2017

Organization of International Events

- **2026**: Local Chair of the Twenty-Seventh ACM Conference on Economics and Computation (EC'26), Rome (Italy).
- **2026**: Organizer of the ACM EC 2026 workshop "Online Learning and Economics" (OLE), July 6 2026, Rome (Italy).
- **2025**: Organizer of the ACM EC 2025 workshop "Online Learning and Economics" (OLE), July 10 2025, Stanford (CA, USA).
- **2025**: Organizer of the Second "Algorithms, Learning, and Games" (ALGA) workshop, June 9-13 2025, Villasimius (Italy).
- **2023**: Organizer of the First "Algorithms, Learning, and Games" (ALGA) workshop, June 5-9 2023, Scicli (Italy).

Organization of Internal Events

- **2025**: Bocconi Theory Day
- **2023**: ELLIS Pre-NeurIPS Fest: poster session of the Milan ELLIS unit held at Bocconi

Invited Talks

- Invited talk for EATCS Prize Ceremony at ICTCS, University of Pescara 2025
- Institut d'Etudes Scienfitiques, "From Matching to Market" workshop, Cargèse 2025
- Politecnico di Milano, FAIR AI Seminar, Milan 2025
- ENSAE, Seminar of the Statistics and Economics Department, Paris 2023
- Criteo's AI Labs, Paris 2023
- Workshop on Game Theory and Machine Learning, London School of Economics 2023
- Workshop "Learning in games", Institute for Mathematical Sciences, Singapore 2023
- London School of Economics Seminar Series on Combinatorics, Games and Optimisation 2022
- ELLIS Milan Artificial Intelligence workshop 2022

- INFORMS session on Operations Research at Facebook 2021
- Invited talk at explAIn workshop (Virtual) 2021
- Facebook London ML Seminars (Virtual) 2021
- Facebook Economics, Algorithms and Optimization Seminars (Virtual) 2021
- Coveo ML Seminars (Virtual) 2021
- Politecnico di Milano, MAPLE workshop, Milan 2019
- Invited talk at International Conference of the AI*IA 2017

Advising & Mentoring

PostDocs

- Martino Bernasconi (2023 – Present)
- Riccardo Poiani (2024 – Present)
- Sarah Sachs (2024 – 2025)

M.Sc. Students

- Annalisa Barbara (2025)
- Emanuele Coccia (2024 – 2025)
- Antonio Preiti (2024)
- Davide Drago (2023)

Teaching at Bocconi

- Deep Learning and Reinforcement Learning (M.Sc. class), Course Director (teaching RL section)
  Spring 2024, 2025, 2026
- Machine Learning (B.Sc. class), Course Director
  Spring 2023, 2024, 2025, 2026
- Optimization (Ph.D. in Statistics and Computer Science), Course Director
  Fall 2021, 2022, 2023, 2024, 2025
- Algorithms (M.Sc. class), Professor
  Spring 2022
- Computer Programming and Database Systems (M.Sc. class), Professor
  Fall 2021, 2022

Prior Teaching

- Game Theory, Politecnico di Milano (M.Sc.), TA 2019–2020
- Informatica A, Politecnico di Milano (B.Sc.), TA '18–'19, '19–'20
- Economics and Computation, Politecnico di Milano (M.Sc.), TA '17–'18, '18–'19

Industry Training & Courses

- Race Strategies Computation, Ferrari S.p.A. 2019

Service

- **Grant peer review**: reviewer for European Research Council (ERC) grant, Vienna Science and Technology Fund (WWTF)
- **Thesis referee/committee member:** Luca Carminati (Politecnico di Torino, 2025), Lukas Zierahn (Politecnico di Torino, 2025), David Sychrovský (Charles University, 2025)
- **Area Chair**: ICML 2026, NeurIPS 2026

- **Program committee member**: ICLR 2026, EC (2025,2026), NeurIPS (2020–2025), ICML (2022, 2023, 2024), AAAI (2018–2023), ECAI (2020), IJCAI (2020–2023), WINE (2021, 2026)
- **Journal peer review**: Transactions on Economics and Computation (2026), Autonomous Agents and Multi-Agent Systems (2026), Mathematical Programming (2025), European Journal of Operational Research (2024), Mathematics of Operations Research (2023, 2025), Journal of Economic Theory (2022, 2026), Management Science (2022)
- **Conference peer review**: ICALP (2026), STOC (2026), FOCS (2025,2026), SODA (2023), EC (2021, 2022), ICML (2021), TheWebConf (2020), AAMAS (2017), IJCAI (2017–2019)

Public Engagement & Outreach

- Speaker at MEETmeTONIGHT (outreach event in Milan within the EU project MEET (Mission: Engagement and Education for Tomorrow)) Milan, 2025
- General-audience article (in Italian): "*Che succede quando i sistemi di IA interagiscono tra loro?*" Eco Magazine, 7/2024 2024